

Document Information

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Definition of Terms, Acronyms and Abbreviations

Term	Description
ASP	Active Server Pages
DD	Design Specification
UI	User Interface
DB	Database
ERD	Entity Relationship Diagram

1 Introduction

1.1 Purpose of Document

This document provides a detailed design specification for the **Refugee Relief Camp Management System**. It describes the system architecture, database design, and application design required to implement the system.

The intended audience includes:

- Developers
- Project supervisors
- Stakeholders (NGOs, camp administrators)

The system follows an Object-Oriented Design Methodology, ensuring modularity, reusability, and scalability.

1.2 Intended Audience

- Software developers (implementation)
- Database designers
- Project evaluators (teachers)
- Camp administrators (non-technical users)

1.3 Document Convention

- Font: Times New Roman
- Size: 12 (body), 14 (headings)
- Bold headings and structured numbering

1.4 Project Overview

The system is designed to digitally manage relief camps, replacing manual processes.

Core functionalities:

- Refugee registration & tracking
- Shelter allocation & capacity management
- Volunteer & doctor management
- Medical records handling
- Inventory tracking
- Donation management
- Incident reporting

The design uses:

- Relational database (ERD-based)
- Layered architecture (UI + Logic + DB)

1.5 Scope

In Scope

- Refugee, volunteer, and doctor registration
- Shelter and camp management

- Medical record system
- Inventory and donations tracking
- Incident reporting

Out of Scope

- AI-based analytics
- GPS tracking
- Government ID integration
- Advanced reporting dashboards

2 Design Considerations

2.1 *Assumptions and Dependencies*

- Users have basic computer literacy
- Internet or local system is available
- Database server is accessible
- NGOs provide accurate data

Dependencies:

- Database system (Oracle SQL)
- Development platform (web)

2.2 *Risks and Volatile Areas*

Risks

- Data inconsistency due to manual input
- System misuse by untrained users
- Limited infrastructure in camps

Volatile Areas

- Changing requirements from NGOs
- Scaling to multiple camps
- Data privacy concerns

Mitigation

- Input validation
- Modular design
- Secure database handling

3 System Architecture

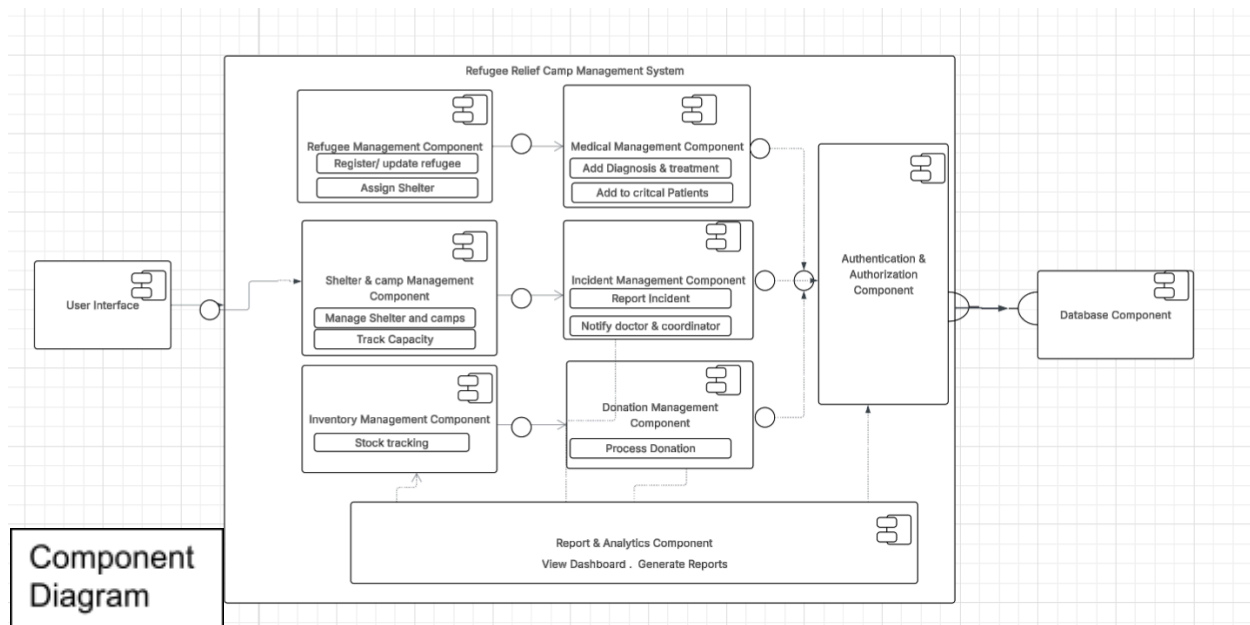
3.1 System Level Architecture

The system is divided into the following components:

1. **User Interface Module**
 - o Forms for data entry
 - o Dashboard for viewing records
2. **Application Logic Module**
 - o Handles business logic
 - o Processes data (assign shelter, record incident, etc.)
3. **Database Module**
 - o Stores all system data
 - o Maintains relationships (ERD-based)

Relationships

- UI interacts with Logic Layer
- Logic Layer interacts with Database
- Database stores structured relational data



3.2 Software Architecture

User Interface Layer

- Web-based user interface (UI)
- Interfaces for:
 - o Coordinator (Admin dashboard)
 - o Doctor (Medical interface)
 - o Volunteer (Incident reporting interface)
 - o Organization/Donor (Donation interface)
- Input forms:

- Refugee Registration
- Volunteer Registration
- Doctor Registration
- Displays:
- Reports (refugee data, inventory, incidents, donations)
- System notifications and confirmations

Application Layer (Business Logic)

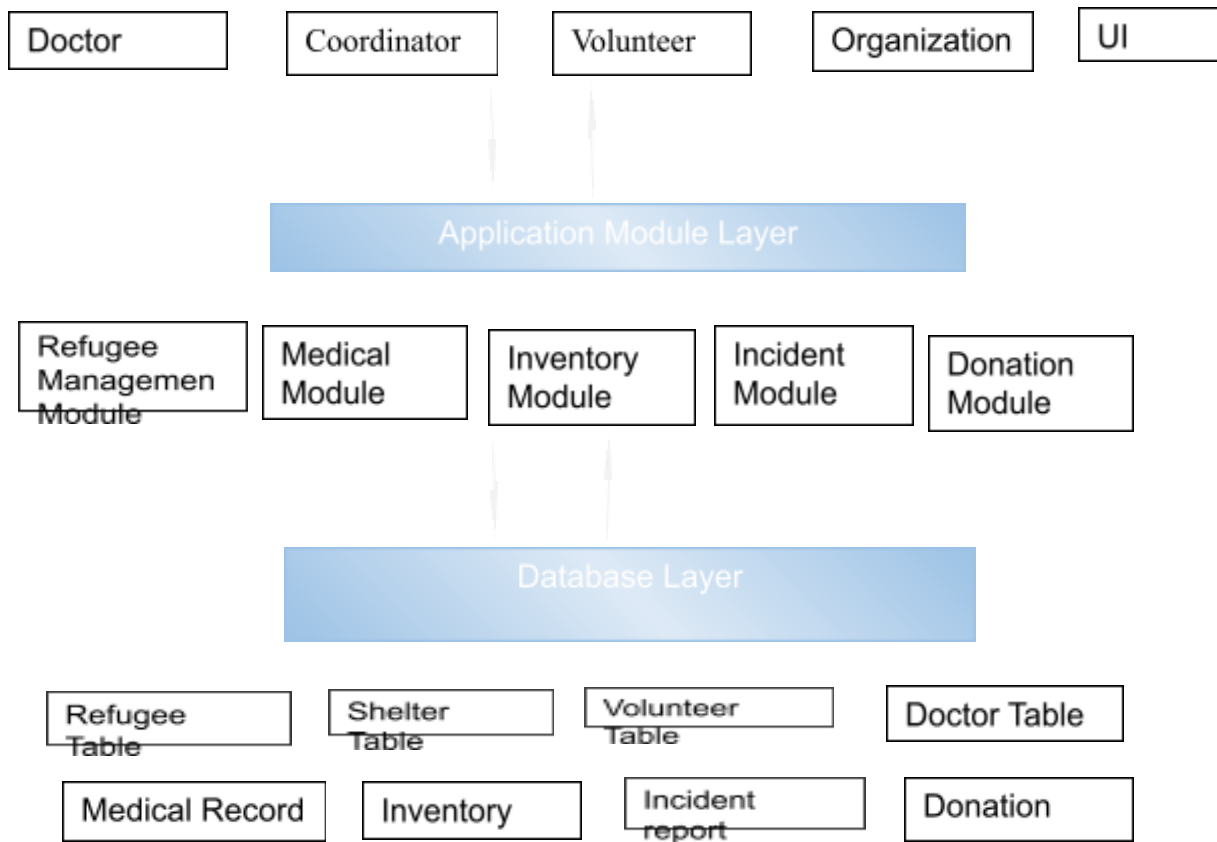
- Handles core system functionality:
- Refugee Management (registration, updates, shelter assignment)
- Medical Management (diagnosis, treatment records)
- Incident Management (reporting and tracking incidents)
- Inventory Management (stock updates and monitoring)
- Donation Management (recording and tracking donations)
- Authentication & Authorization
- Performs:
- Data validation
- Business rule enforcement
- Processing user requests

Data Access Layer

- SQL queries
- CRUD operations
- Manages database tables:
 - Refugee
 - Shelter
 - Volunteer
 - Doctor
 - Medical_Record
 - Inventory
 - Incident_Report
 - Organization
 - Donations



User Interface layer



4 Design Strategy

Future Extension

- Can support multiple shelters
- Can integrate mobile apps

System Reuse

- Modular components (registration, inventory) reusable

User Interface

- Simple forms

Data Management

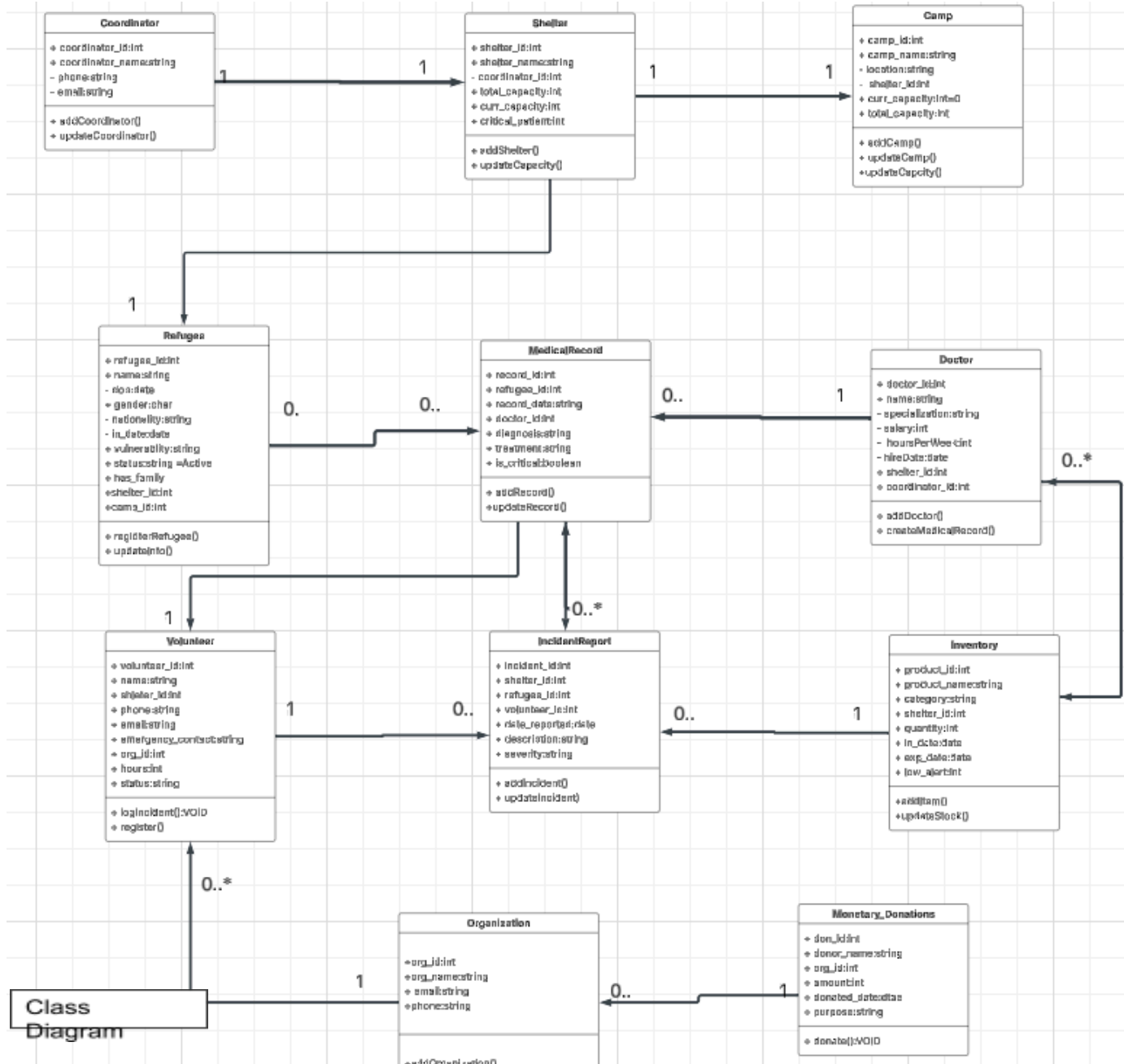
- Centralized relational database
- Persistent storage

Concurrency

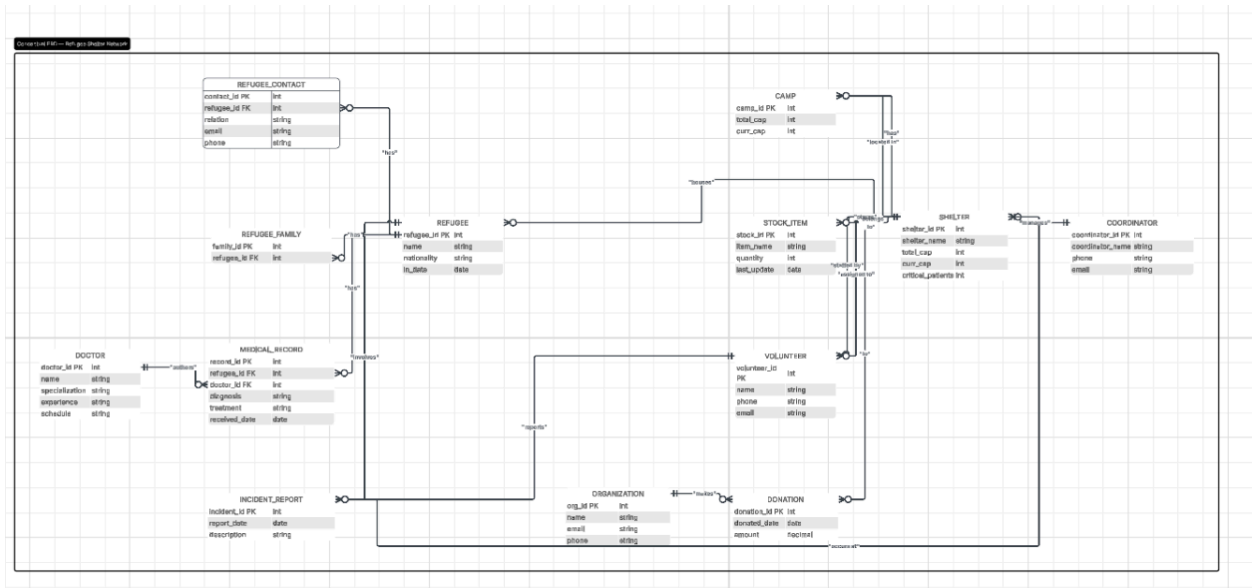
- Multiple users can access system
- Controlled DB transactions

5 Detailed System Design

5.1 Database Design



5.1.1 ER Diagram



5.1.2 Data Dictionary

5.1.2.1 Data 1

< Data 1 >						
Name	Refugee					
Alias	ref					
Where-used/how-used	Used in registration, shelter assignment, medical records, and incident reports					
Content description	Refugee = Refugee_id + Name + In_date + Vulnerability + Has_Family					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>Refugee_id</i>	<i>Unique Refugee id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Name</i>	<i>Refugee name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>In_date</i>	<i>Date of arrival</i>	<i>DATE</i>		<i>No</i>		
<i>Vulnerability</i>	<i>Health/priority Status</i>	<i>VARCHAR</i>	<i>20</i>	<i>No</i>		
<i>Shelter_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>Yes</i>	<i>Null</i>	<i>FK</i>

<i>Has_Family</i>	<i>Family presence</i>	<i>BOOLEAN</i>		<i>Yes</i>	<i>False</i>	
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5.1.2.2 Data 2

< Data 2>						
Name	Shelter					
Alias	sh					
Where-used/how-used	Used to assign refugees and manage capacity					
Content description	Shelter = Shelter_id + Shelter_name + Total_capacity + Curr_capacity + Coordinator id					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>Shelter_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Shelter_name</i>	<i>Shelter name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Coordinator_id</i>	<i>Assigned coordinator id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Total_capacity</i>	<i>Maximum capacity</i>	<i>INT</i>		<i>No</i>		
<i>Cuurent_capacity</i>	<i>Current occupancy</i>	<i>INT</i>		<i>Yes</i>		
<i>Critical_Patients</i>	<i>Critical patients presence</i>	<i>BOOLEAN</i>		<i>Yes</i>	<i>False</i>	

5.1.2.3 Data 3

< Data 3>	
Name	Camp
Alias	C
Where-used/how-used	Links shelter and capacity tracking
Content description	Camp = Camp_id + Shelter_id + Total_capacity + Curr_capacity

Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Camp_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Shelter_id</i>	<i>Shelter id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Total_capacity</i>	<i>Maximum capacity</i>	<i>INT</i>		<i>No</i>		
<i>Cuurent_capacity</i>	<i>Current occupancy</i>	<i>INT</i>		<i>Yes</i>		

5.1.2.4 Data 4

< Data 4>						
Name	Coordinator					
Alias	co					
Where-used/how-used	Manages shelters					
Content description	Coordinator = Coordinator_id + Name + Phone + Email					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Coordinator_id</i>	<i>Unique Coordinator id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Name</i>	<i>Coordinator name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Phone</i>	<i>Contact Number</i>	<i>VARCHAR</i>	<i>11</i>	<i>No</i>		
<i>Email</i>	<i>Email Address</i>	<i>VARCHAR</i>	<i>50</i>	<i>Yes</i>		

5.1.2.5 Data 5

< Data 5>	
Name	Volunteer
Alias	v
Where-used/how-used	Used in incident reporting and shelter support

Content description		Volunteer = Volunteer_id + Name + Phone + Email + Shelter_id				
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Volunteer_id</i>	<i>Unique Volunteer id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Name</i>	<i>Volunteer name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Shelter_id</i>	<i>Assigned shelter id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Org_id</i>	<i>Organization id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Phone</i>	<i>Contact number</i>	<i>VARCHAR</i>	<i>11</i>	<i>Yes</i>		
<i>Email</i>	<i>Email Address</i>	<i>VARCHAR</i>	<i>50</i>	<i>Yes</i>	<i>False</i>	
<i>Emergency_contact</i>	<i>Emergency Contact</i>	<i>VARCHAR</i>	<i>11</i>	<i>No</i>		
<i>Hours</i>	<i>Number of hours</i>	<i>INT</i>		<i>No</i>	<i>0</i>	

5.1.2.6 Data 6

< Data 6>						
Name	Doctor					
Alias	doc					
Where-used/how-used	Used in medical record					
Content description	Doctor = Doctor_id + Name + Specialization + Salary + HireDate					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Doctor_id</i>	<i>Unique doctor id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Name</i>	<i>Doctor name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Specialization</i>	<i>Field</i>	<i>INT</i>		<i>No</i>		

<i>HoursPerWeek</i>	<i>Hours logged per week</i>	<i>INT</i>		<i>No</i>		
<i>Salary</i>	<i>Salary</i>	<i>FLOAT</i>		<i>Yes</i>		
<i>Hire_date</i>	<i>Joining date</i>	<i>DATE</i>		<i>Yes</i>	<i>False</i>	

5.1.2.7 Data 7

< Data 7>						
Name	Medical_Record					
Alias	Mr					
Where-used/how-used	Stores Refugee medical data					
Content description	Medical_Record = Record_id + Refugee_id + Doctor_id + Diagnosis + Treatment					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Record_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Doctor_id</i>	<i>Shelter name</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Refugee_id</i>	<i>Assigned coordinator id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Record_date</i>	<i>Maximum capacity</i>	<i>INT</i>		<i>No</i>		
<i>Diagnoses</i>	<i>Current occupancy</i>	<i>TEXT</i>		<i>No</i>		
<i>Treatment</i>	<i>Critical patients presence</i>	<i>TEXT</i>		<i>Yes</i>	<i>False</i>	

5.1.2.8 Data 8

< Data 8>	
Name	Inventory

Alias		inv				
Where-used/how-used		Track resources in shelters				
Content description		Inventory = Product_id + Product_name + Quantity + Shelter_id				
Column Name	Description	Type	Length	Null able	Default Value	Key Type
Product_id	Unique Product id	INT		no		PK
Product_name	Product name	VARCHAR	50	No		
Shelter_id	Assigned shelter id	INT		No		FK
Quantity	Available stock	INT		No		
In_date	In date	DATE		Yes		
Expiry Date	Expiry date	DATE		Yes		

5.1.2.9 Data 9

< Data 9>						
Name		Incident_Report				
Alias		inc				
Where-used/how-used		Record issues in camp				
Content description		Incident_Report = Incident_id + Refugee_id + Volunteer_id + Description + Date				
Column Name	Description	Type	Length	Null able	Default Value	Key Type
Incident_id	Unique incident id	INT		no		PK
Shelter_id	Shelter id	INT		No		FK
Refugee_id	Assigned refugee id	INT		No		FK
Volunteer_id	Volunteer id	INT		No		FK

<i>Date</i>	<i>Incident date</i>	<i>DATE</i>		<i>No</i>		
<i>Description</i>	<i>Incident detail</i>	<i>TEXT</i>		<i>No</i>		

5.1.2.10 Data 10

< Data 10>						
Name	Organization					
Alias	org					
Where-used/how-used	Linked to donations					
Content description	Organization = Org_id + Name + Phone + Email					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Org_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Name</i>	<i>Shelter name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Phone</i>	<i>Assigned coordinator id</i>	<i>VARCHAR</i>	<i>11</i>	<i>No</i>		
<i>Email</i>	<i>Maximum capacity</i>	<i>VARCHAR</i>	<i>50</i>	<i>Yes</i>		

5.1.2.11 Data 11

< Data 11>						
Name	Monetary_donation					
Alias	don					
Where-used/how-used	Tracks funding					
Content description	Donations = Donation_id + Org_id + Amount + Date					
Column Name	Description	Type	Length	Null able	Default Value	Key Type

<i>Donor_id</i>	<i>Unique Shelter id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Donor_name</i>	<i>Shelter name</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Org_id</i>	<i>Assigned coordinator id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Amount</i>	<i>Maximum capacity</i>	<i>INT</i>		<i>No</i>		
<i>Donated_date</i>	<i>Current occupancy</i>	<i>DATE</i>		<i>No</i>		

5.1.2.12 Data 12

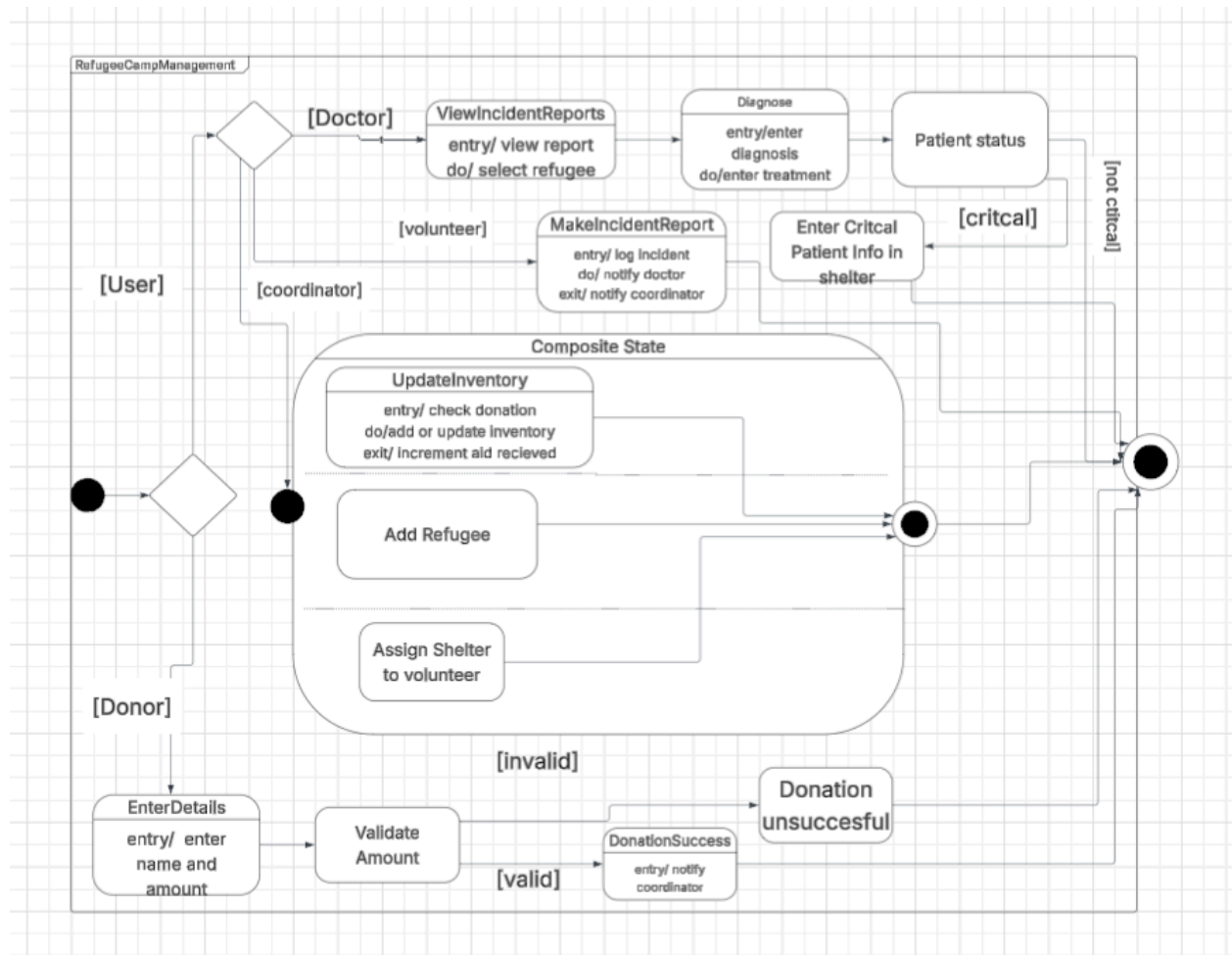
< Data 12>						
Name	Refugee_contact					
Alias	rcon					
Where-used/how-used	Used to record refugee contact outside of the shelter					
Content description	Refugee_contact = Contact_id + Refugee_id+ relation + Email + Phone					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Contact_id</i>	<i>Unique Contact id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Relation</i>	<i>Relationship to refugee</i>	<i>VARCHAR</i>	<i>50</i>	<i>No</i>		
<i>Refugee_id</i>	<i>Assigned refugee id</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Phone</i>	<i>Contact Number</i>	<i>VARCHAR</i>	<i>11</i>	<i>No</i>		
<i>Email</i>	<i>Email Address</i>	<i>VARCHAR</i>	<i>50</i>	<i>Yes</i>		

5.1.2.13 Data 13

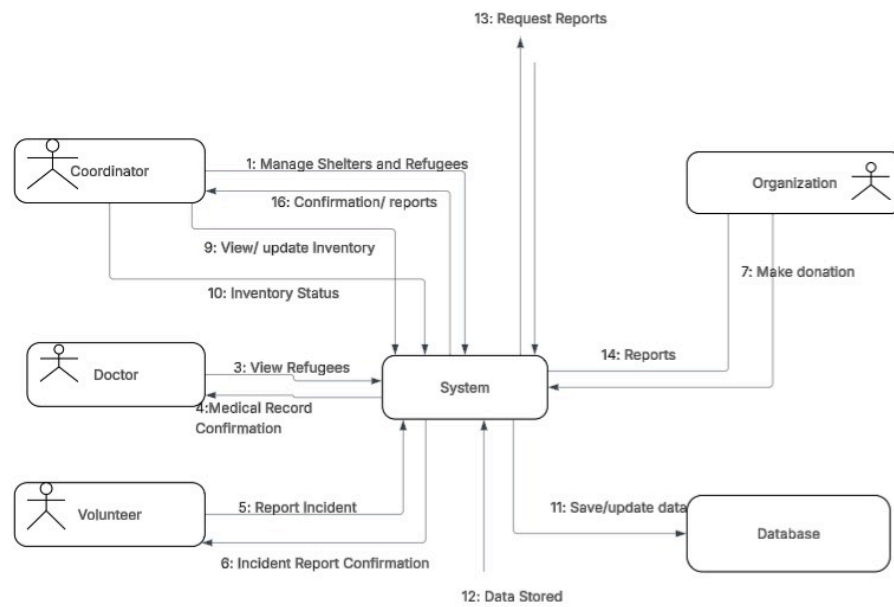
< Data 13>

Name	Refugee_Family					
Alias	rfam					
Where-used/how-used	Used to keep record of refugee family within the shelter					
Content description	Refugee_Family = Family_id + Refugee_id + Ref Head id + Relation					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Family_id</i>	<i>Unique family id</i>	<i>INT</i>		<i>no</i>		<i>PK</i>
<i>Refugee_id</i>	<i>Refugee name</i>	<i>INT</i>		<i>No</i>		<i>FK</i>
<i>Ref_Head_id</i>	<i>Refugee head id</i>	<i>INT</i>		<i>No</i>		
<i>Relation</i>	<i>Relationship with head</i>	<i>VARCHAR</i>	<i>20</i>	<i>No</i>		

Application Design

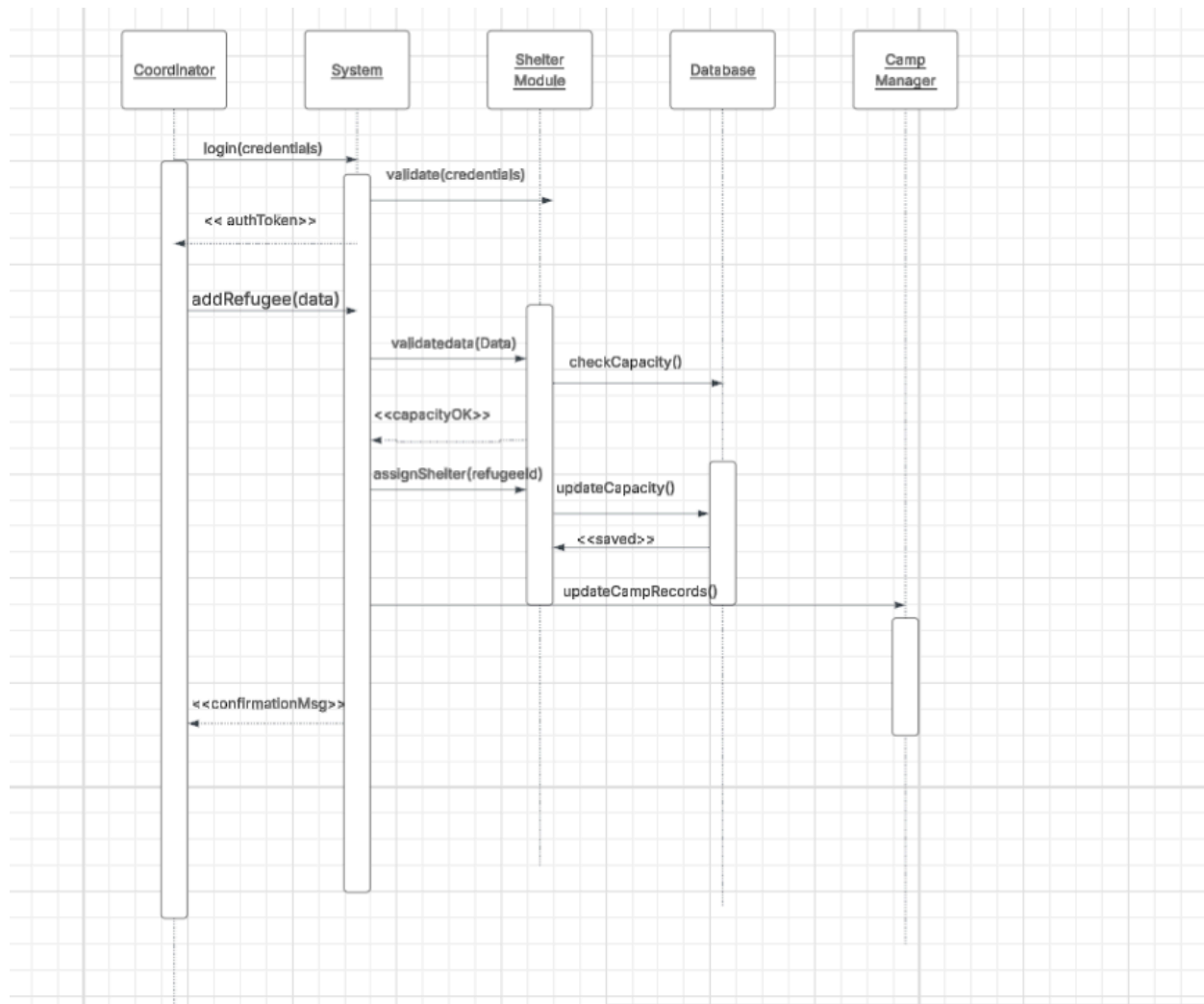


Activity Diagram



5.1.3 Sequence Diagram

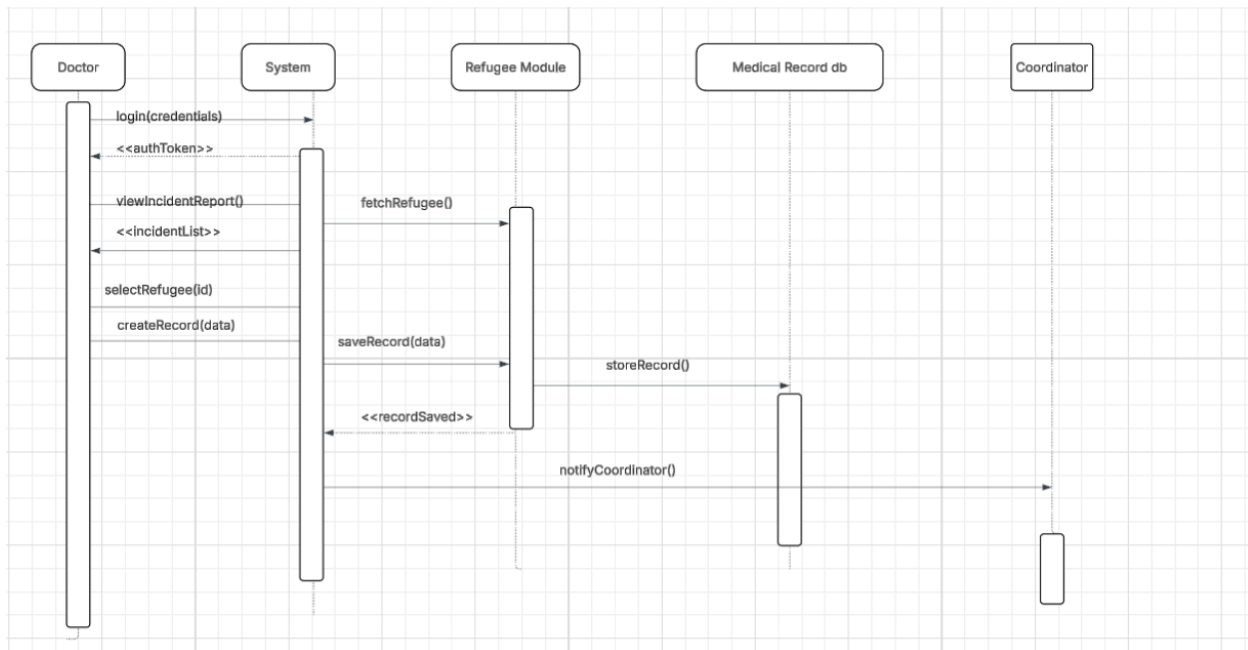
5.1.3.1 <Coordinator registers a refugee >



Flow:

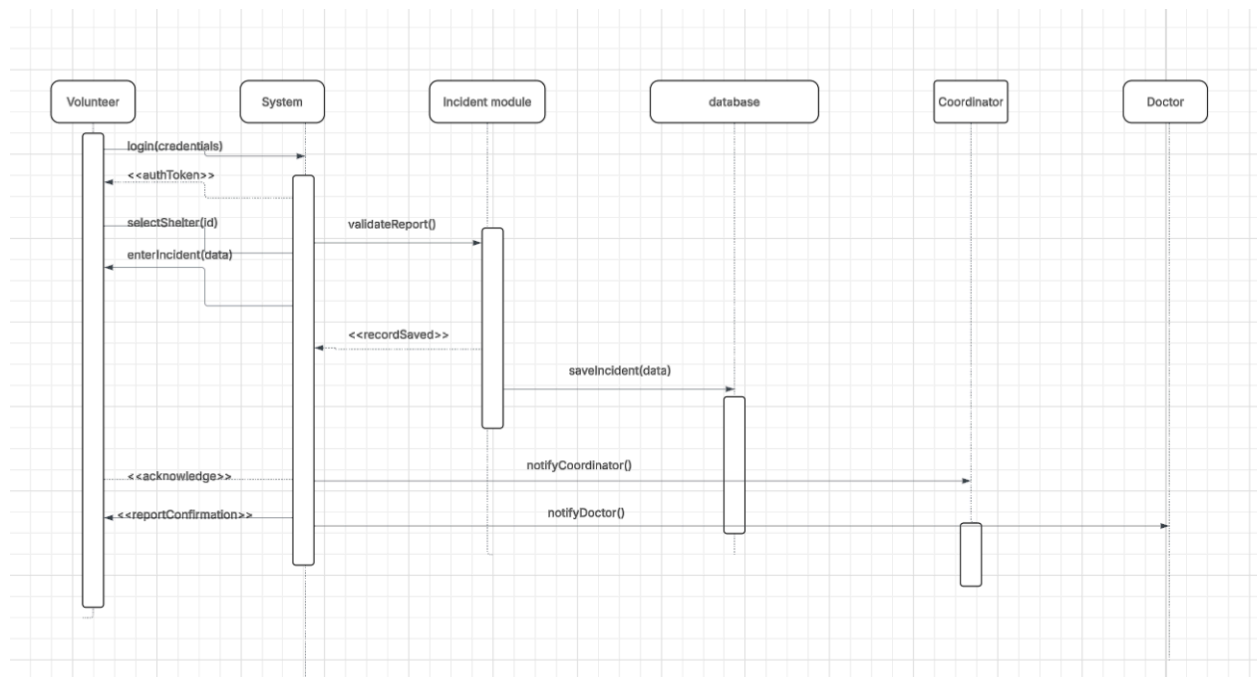
1. System starts (filled black circle)
2. Coordinator logs in
3. Coordinator adds or updates refugee information
4. System validates the data entered
5. System assigns a shelter to the refugee
6. System updates the shelter's current capacity count
7. System saves the data to the database
8. Process ends (circle with dot)

5.1.3.2 <Doctor creates medical record >

**Flow:**

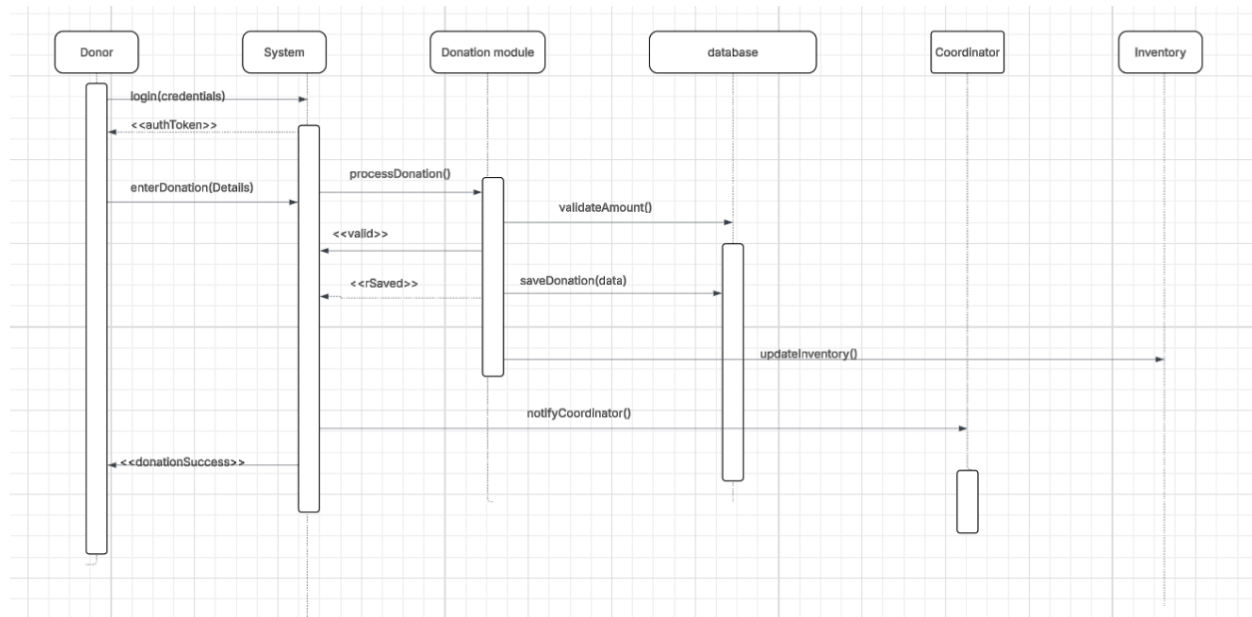
- System starts
- Doctor logs in
- Doctor views the list of incident reports
- Doctor selects a specific refugee from the list
- Doctor creates a new medical record for that refugee
- Doctor adds diagnosis and treatment details
- System saves the record to the database
- Process ends

5.1.3.3 <Volunteer reports an incident>



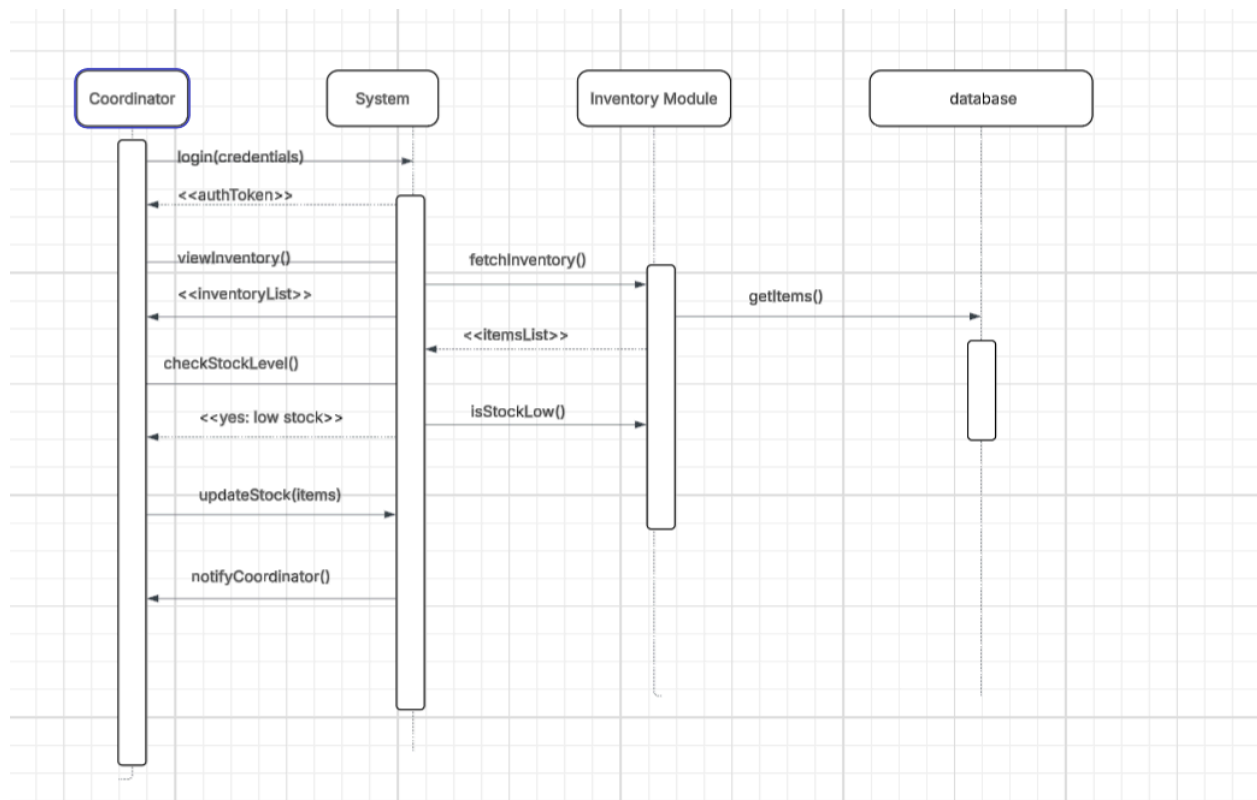
Flow:

1. System starts
2. Volunteer logs in
3. Volunteer selects the refugee and/or shelter involved
4. Volunteer enters the incident details
5. Volunteer submits the report
6. System validates the report
7. System saves the incident record
8. Process ends

5.1.3.4 <Donor makes donation>**Flow:**

- System starts
- Donor logs in
- Donor enters donation details (amount, purpose, etc.)
- System confirms donation successful
- Process ends

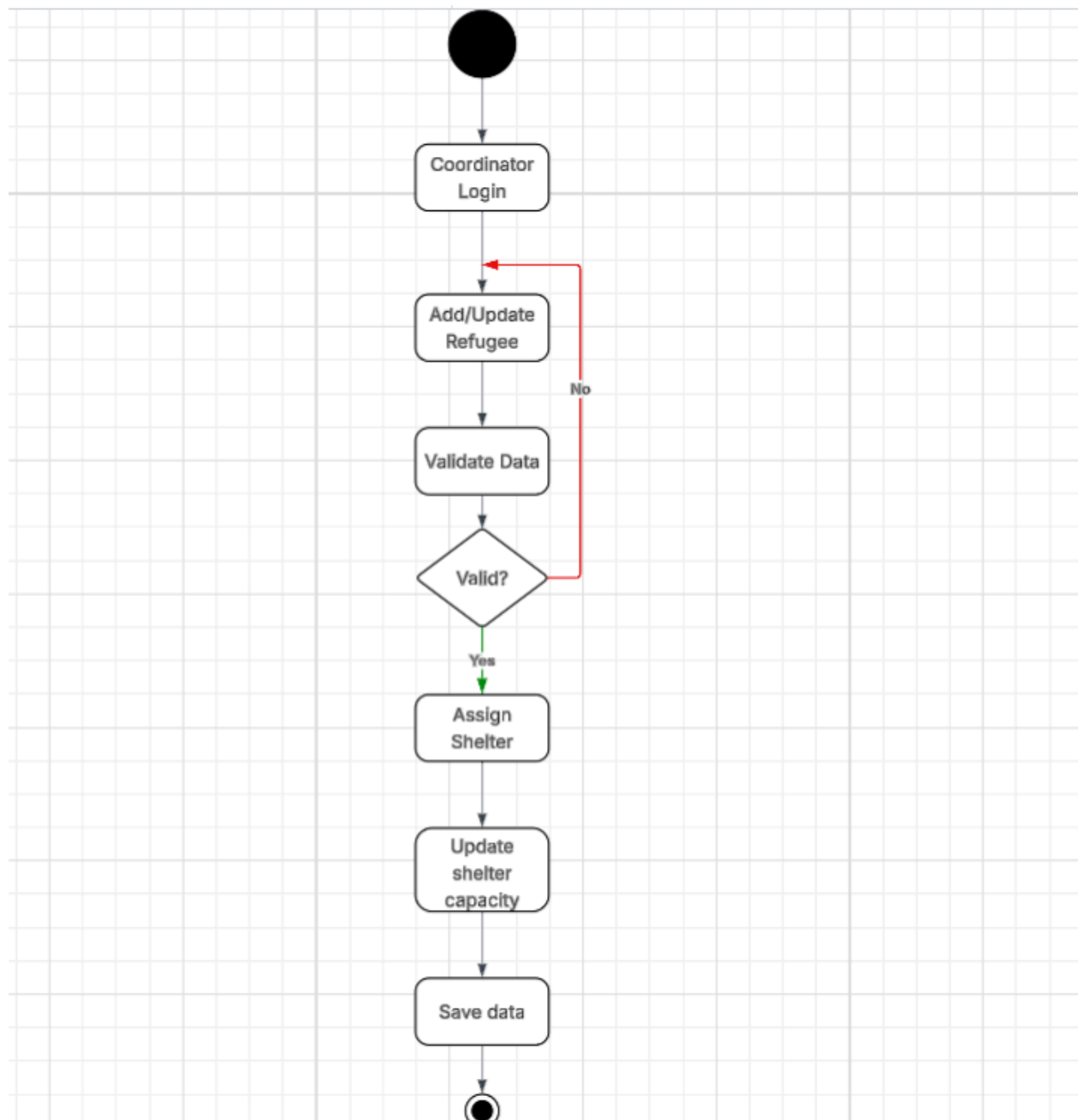
5.1.3.5 <Coordinator manages inventory>

**Flow:**

1. System starts
2. User logs in
3. User views the inventory list
4. System checks current stock levels
5. **Decision: Is stock low?**
 - a. **Yes** → User adds or updates stock quantities → System saves changes → End
 - b. **No** → System goes directly to end (no changes needed)

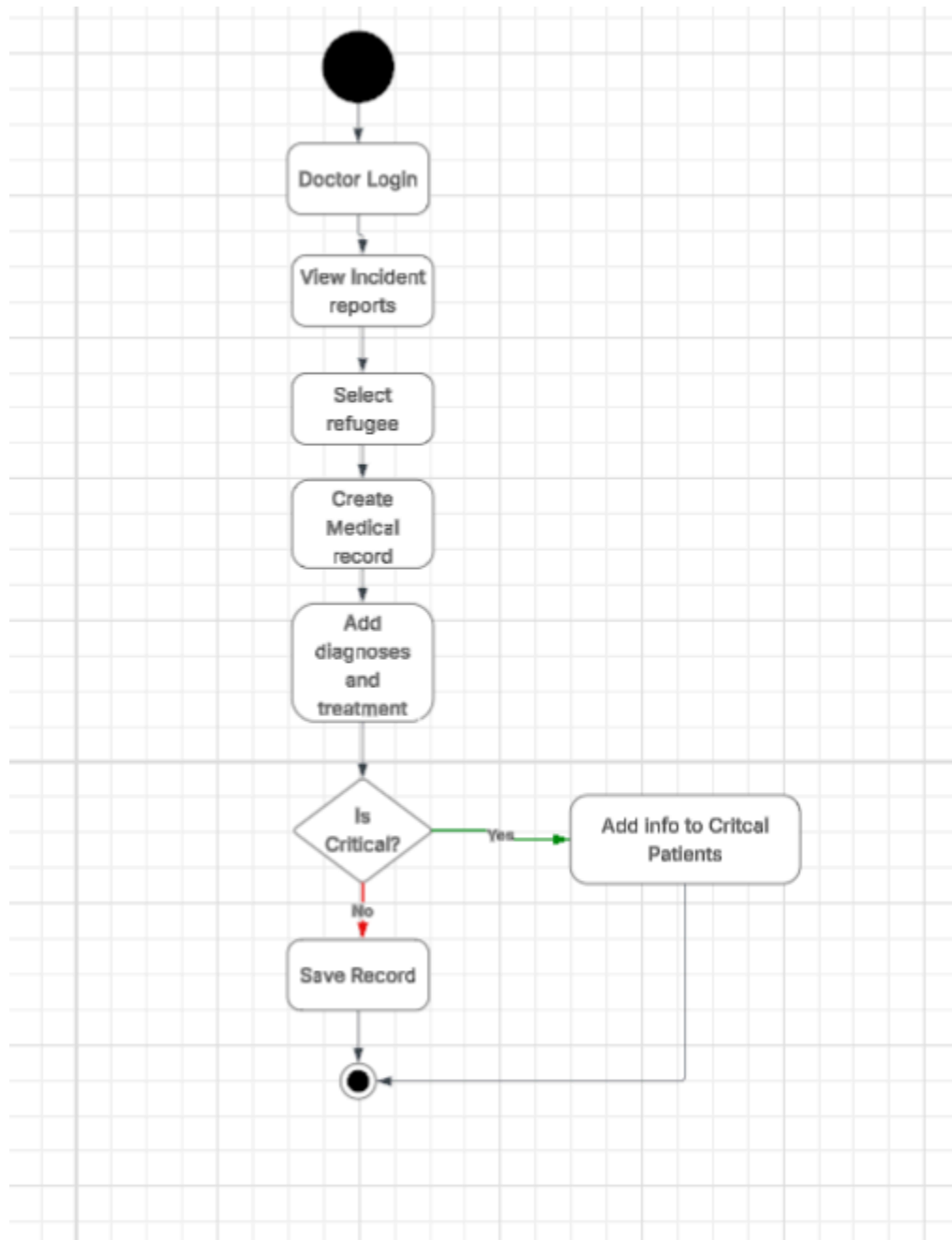
5.1.4 State Diagram

5.1.4.1 <Coordinator + System>



The process begins when the coordinator logs into the system. After login, they either add a new refugee or update an existing one by entering personal details like name, nationality, gender, and vulnerability status. The system then validates all the entered data to make sure nothing is missing or incorrect. Once validated, the system assigns an appropriate shelter to the refugee based on availability. The shelter's current capacity count is then automatically updated to reflect the new addition. Finally, all data is saved to the database and the process ends.

5.1.4.2 <Doctor>

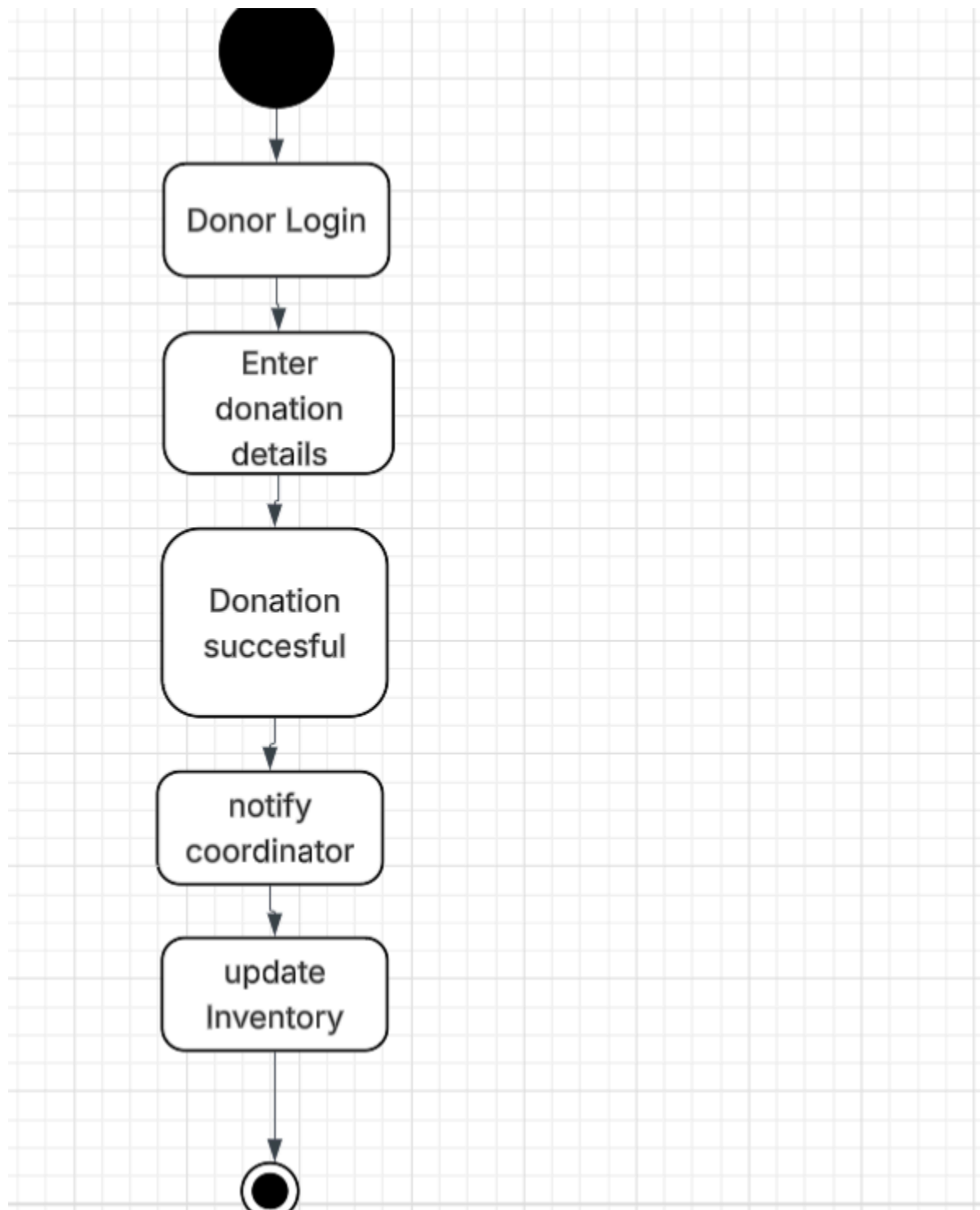


The doctor logs into the system. Instead of directly searching for a refugee, the doctor first views the list of incident reports — meaning medical records are created in response to reported incidents. The doctor then selects the specific refugee from that incident list. A new medical record is created for that refugee, and the doctor fills in the diagnosis and treatment details. The completed record is then saved to the database.

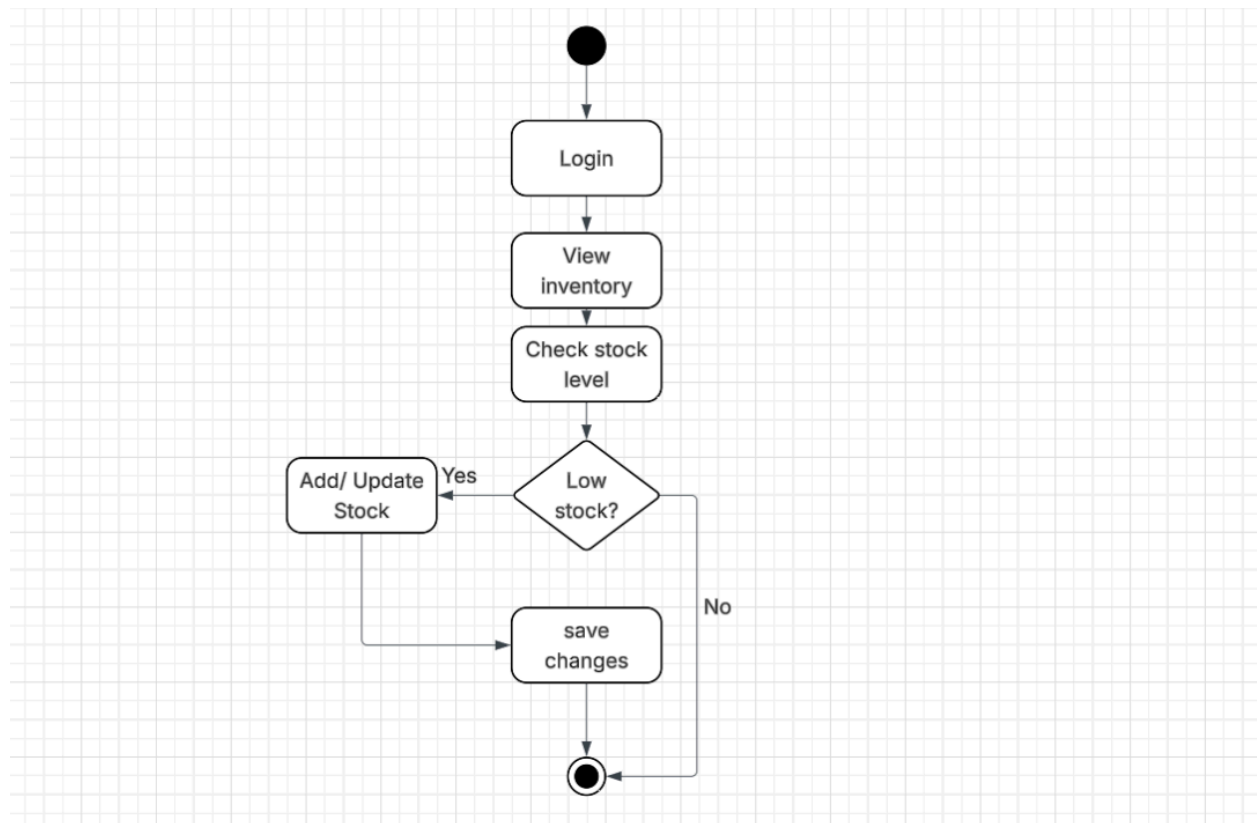
5.1.4.3 <Volunteer>



The volunteer logs in and selects the refugee or shelter that the incident involves. They then enter the full details of the incident — description, severity, date. The report is submitted and the system validates it to check all required fields are filled. Once validated, the incident is saved to the database.

5.1.4.4 <Donor>

The donor logs into the system and enters the donation details such as amount, purpose, and organization information if any. The system processes and confirms the donation as successful, updating inventory, notifying the coordinator, and sending a receipt and the process ends.

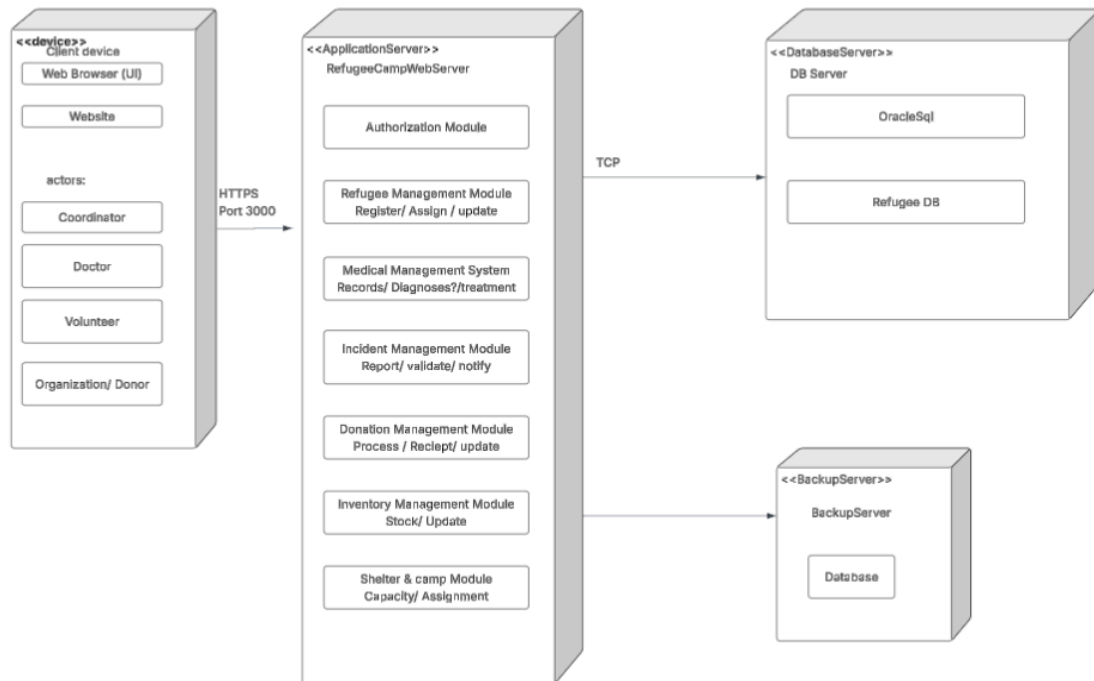
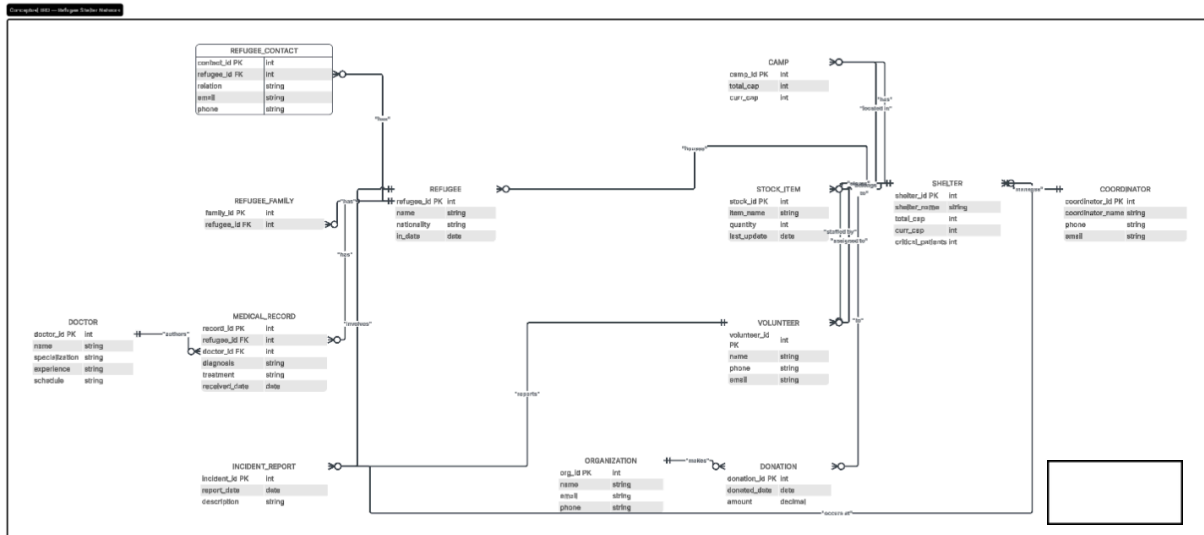
5.1.4.5 <Coordinator + inventory manager>

The coordinator logs in and views the current inventory list. The system checks the stock level of all items. A decision is made — if stock is low, the coordinator adds or updates the stock quantities and the changes are saved. If stock is not low, no changes are needed and the process ends immediately.

References

- 📖 Course material (Software design analysis, Object Oriented programming)
- Database design concepts (ER Modeling)

6 Appendices



Deployment
diagram